

Name.....

Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST-MARCH 2023

SIXTH SEMESTER

CST 302 –COMPILER DESIGN

Time: 2 Hours

Max. Marks: 50

PART A

Answer **all** questions (3x5=15 marks)

Qn No.	Questions	Marks	K level	CO & KL
1	Identify the lexemes that make up the tokens in the following program <pre>int max (i , j) int i , j ; /*return maximum of integers i and j */ { return i > j ? i : j ; }</pre>	3	K3	CO1 & K3
2	Explain the importance of sentinels in input buffering used in lexical analysis.	3	K2	CO1 & K3
3	Consider the grammar , bexpr -> bexpr or bterm bterm bterm -> bterm and bfactor bfactor bfactor -> not bfactor (bexpr) true false Construct a parse tree for the sentence not (true or false)	3	K3	CO2 & K3
4	Is the grammar $S \rightarrow S (S) \epsilon$ ambiguous? Illustrate your answer.	3	K3	CO2 & K3
5	Compare top down parser and bottom up parser.	3	K2	CO3 & K3

PART B

Answer **any five** questions (7x5=35 marks)

Qn. No 6,7 and 8 – Answer any 2

Qn. No 9,10 and 11 – Answer any 2

Qn. No 12 is mandatory

Qn No.	Questions	Marks	K level	CO & KL
6	With a neat diagram explain the phases of a compiler for the input position := initial + rate*60 with the variables declared as float.	7	K3	CO1 & K3
7	Draw the transition diagram for identifier, relational operators and number.	7	K3	CO1 & K3

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8	a) Explain about tokens, patterns and lexemes.	3	K1	CO1& K3
	b) How will you specify and recognize tokens using regular expressions?	4	K1	CO1& K3
9	Find out the FIRST () and FOLLOW () of all the non-terminals in following grammars : a) $S \rightarrow AaBb \mid BbBa$ $A \rightarrow \epsilon$ $B \rightarrow \epsilon$	3.5	K3	CO2 & K3
	b) $S \rightarrow 1AB \mid \epsilon$ $A \rightarrow 11AC \mid 0C$ $B \rightarrow 0S$ $C \rightarrow 1$	3.5	K3	CO2 & K3
10	a) Depict the steps in removing left recursion in the following grammar: $S \rightarrow (L) \mid a$ $L \rightarrow L, S \mid S$	3	K3	CO2& K3
	b) Construct a predictive parsing table for the above grammar.	4	K3	CO2& K3
11	Explain the working of a bottom up parser with the help of an example.	7	K2	CO3& K3
12	Design a recursive descent parser without backtracking for the grammar , $S \rightarrow c A d$ $A \rightarrow ab/b$	7	K3	CO2 & K2



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE SECOND INTERNAL TEST - June 2023

SIXTH SEMESTER

CST 302 –COMPILER DESIGN

Time: 2 Hours

Max. Marks: 50

PART A

Answer *all* questions (3x5=15 marks)

Qn No.	Questions	Marks	K level	CO & KL
1	What is handle pruning in bottom up parsing?	3	K2	CO3 & K3
2	What are the parsing conflicts in LR(0)?	3	K2	CO3 & K3
3	Construct the syntax tree and then draw the DAG for the statement: $e := (a*b) + (c-d) * (a*b)$	3	K3	CO4 & K3
4	Differentiate between synthesized attributes and inherited attributes with an example.	3	K2	CO4 & K3
5	Explain about Loop-Invariant Code Motion with example.	3	K2	CO5 & K3

PART B

Answer *any five* questions (7x5=35 marks)

Qn. No 6,7 and 8 – Answer any 2

Qn. No 9,10 and 11 – Answer any 2

Qn. No 12 is mandatory

Qn No.	Questions	Marks	K level	CO & KL
6	Construct LALR parse table for the grammar: $A \rightarrow BB$ $B \rightarrow bB \mid d$	7	K3	CO3 & K3
7	Construct the canonical LR(0) items of the following grammar $S \rightarrow AA$ $A \rightarrow aA \mid b$	7	K3	CO3 & K3
8	Write the SDD for a desk calculator and draw the annotated parse tree for the expression: $4 * 5 + 6 - (3 * 2)$	7	K3	CO4 & K3
9	Explain bottom- up evaluation of s-attributed definitions.	7	K2	CO4 & K3
10	Construct the DAG and three address code for the expression $a+a*(b-c)+b*(b-c)+b$	7	K3	CO4 & K3

car
low
low
do

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11	Write Three-address code, Triples, Quadruples, Indirect triples for the expression $S: = A - B + C * D - E + F$	7	K3	CO4 & K3
12	Explain about principal sources of optimization in Compiler Design.	7	K2	CO5 & K3

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Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R, S) / S6 (PT) (R) Examination June 2023 (2019 Scheme)

Course Code: CST302**Course Name: COMPILER DESIGN**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|--|--|-----|
| 1 ✓ | What is the relevance of input buffering in lexical analysis? | (3) |
| 2 ✓ | Draw a transition diagram to represent relational operators. | (3) |
| 3 ✓ | With an example write the steps to remove left recursion. | (3) |
| 4 ✓ | What is left factoring? Left factor the following grammar, C | (3) |
| $E \rightarrow E + T \mid T$ $T \rightarrow \text{float} \mid \text{float} * T \mid (E)$ | | |
| 5 ✓ | Differentiate CLR and LALR parsers. | (3) |
| 6 ✓ | What are the possible actions of a shift-reduce parser? | (3) |
| 7 ✓ | Convert the expression $a = b * -c + b * -c$ into quadruple? | (3) |
| 8 ✓ | Define SDD with an example. | (3) |
| 9 ✓ | Explain common sub expression elimination with an example. | (3) |
| 10 ✓ | How the peephole optimization technique makes its role in the compilation process? | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

- 11 a) What are the various phases of a compiler? Explain each phase in detail by using the input statement. $\text{position} := \text{initial} + \text{rate} * 60$ (8)
- b) Differentiate tokens, patterns and lexemes with the help of an example. (6)

OR

- 12 a) Write short notes on compiler construction tools. (6)
- b) Explain in detail about buffer pairs and sentinels. (8)

Module II

- 13 a) Find FIRST set and FOLLOW set of each nonterminal in the following grammar. (6)

$$S \rightarrow aBDh \mid bBc$$

$$B \rightarrow eC$$

$$C \rightarrow bC \mid \epsilon$$

$$D \rightarrow EF$$

$$E \rightarrow g \mid \epsilon$$

$$F \rightarrow f \mid \epsilon$$

- b) Explain the error recovery strategies in parsing. (8)

OR

- 14 a) i) Show that the given grammar is ambiguous or not. (7)

$$E \rightarrow E + E \mid E - E$$

$$E \rightarrow E * E \mid E / E$$

$$E \rightarrow E \wedge E$$

$$E \rightarrow (E) \mid id$$

Also eliminate ambiguity from the above grammar.

(Precedence order: id, (, ^, * and /, + and -)

- b) Construct a non-recursive predictive parsing table for the following grammar: (7)

$$S \rightarrow (L) \mid a$$

$$L \rightarrow L, S \mid S$$

Also prove that the grammar is LL(1) or not.

Module III

- 15 a) What is a shift-reduce parser? Explain in detail the conflicts that may occur during shift-reduce parsing. (8)

- b) Construct canonical LR(1) collection of items for the grammar below: (6)

$$S \rightarrow CC$$

$$C \rightarrow aC$$

$$C \rightarrow d$$

OR

- 16 Consider the grammar (14)

$$S \rightarrow Aa \mid bAc \mid dc \mid bda$$

$$A \rightarrow d$$

Construct a LALR parsing table for the grammar given above. Verify whether the input string "bdc" is accepted by the grammar or not.

Module IV

- 17 a) Define the following terms and give suitable example for each. (6)
 i) Three-address code ii) Triples iii) Quadruples.
 b) Explain static allocation and heap allocation strategies. (8)

OR

- 18 a) Construct a syntax directed definition for an assignment statement. (7)

$S \rightarrow E$

$E \rightarrow E1 + E2$

$E \rightarrow E1 * E2$

$E \rightarrow - E1$

$E \rightarrow (E1)$

$E \rightarrow id$

Also construct an annotated parse tree for the input string: $6 * 8 + 5$.

- b) Generate an intermediate code for the following code segment along with the required syntax-directed translation scheme. (7)

if ($a > b$)

$x = a + b$

else

$x = a - b$

where a and x are of real and b of int type data.

Module V

- 19 a) Explain different code optimization techniques. (8)
 b) Generate a code sequence for the assignment $d = (a - b) + (a - c) + (a - c)$. (6)

OR

- 20 a) Explain the design issues of a code generator. (7)
 b) Illustrate the optimization of basic blocks with examples. (7)
